

Chapter 4

The Central Limit Theorem and General Normal Distributions

Chapter 3 introduced the standard normal distribution $N(0, 1)$: mean zero, standard deviation one. But real measurements rarely have mean zero and standard deviation one. Human heights average around 67 inches with standard deviation 3 inches. Exam scores might have mean 75 and standard deviation 12. Coffee cups are filled with a mean of 8.1 ounces and standard deviation 0.2 ounces.

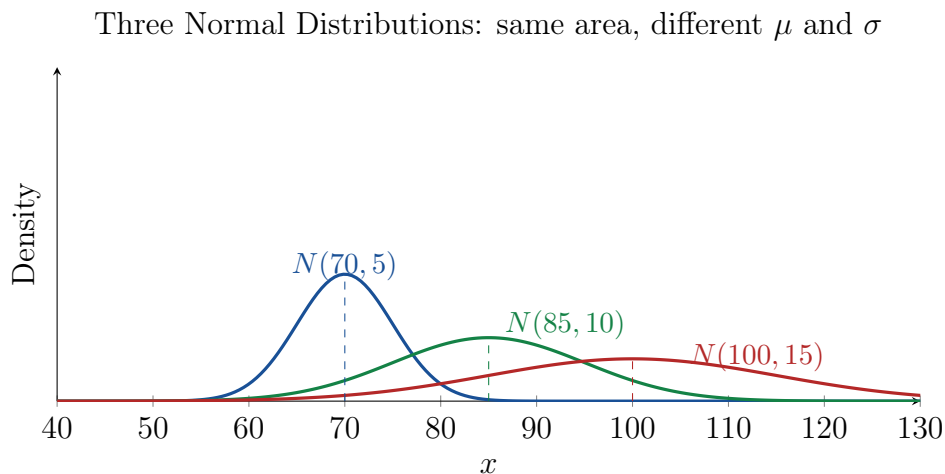
This chapter does two things. First, it extends the normal distribution to any mean μ and standard deviation σ using a single converting formula, the ***z*-score**. Second — and this is the deepest result in the course — it reveals that *sample means themselves* follow a normal distribution, regardless of the shape of the original population. This result, the **Central Limit Theorem**, is the engine that drives every inference procedure from Chapter 5 onward.

4.1 The General Normal Distribution

Definition: General Normal Distribution. A random variable X has a **normal distribution** with mean μ and standard deviation σ , written $X \sim N(\mu, \sigma)$, if its density curve is a bell shape centered at μ with spread controlled by σ .

Changing μ slides the bell left or right along the axis. Changing σ makes the bell wider (larger σ) or narrower (smaller σ). The total area is always 1.

Visual Insight. The three curves below all have the same total area. Only their centers and spreads differ.



The blue curve is tall and narrow ($\sigma = 5$): values are tightly clustered near μ . The red curve is short and wide ($\sigma = 15$): values are spread far from μ . Same area, very different pictures.

4.2 The z -Score: Converting to Standard Units

We have one table: the standard normal table, which works only for $N(0, 1)$. To find probabilities for any $N(\mu, \sigma)$, we convert the question to standard units.

Definition: z -Score. The z -score of a value x from a $N(\mu, \sigma)$ distribution is:

$$z = \frac{x - \mu}{\sigma}.$$

It measures how many standard deviations x is above (positive) or below (negative) the mean.

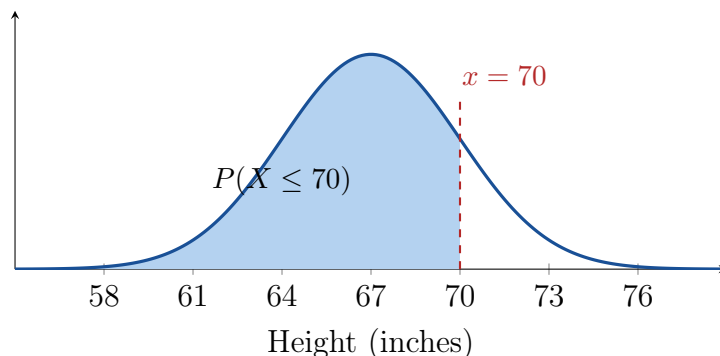
Key Formula. To find $P(X \leq x)$ where $X \sim N(\mu, \sigma)$:

1. Compute $z = \frac{x - \mu}{\sigma}$.
2. Look up $P(Z \leq z)$ in the standard normal table.

For an interval: convert both endpoints to z -scores, then subtract table values.

Visual Insight. The z -score does not change the shape or the probability. It rescales the axis so that the mean maps to 0 and one standard deviation maps to 1. The shaded area stays exactly the same before and after the rescaling.

$X \sim N(67, 3)$: shaded region is $P(X \leq 70)$



Converting: $z = \frac{70 - 67}{3} = 1.00$. So $P(X \leq 70) = P(Z \leq 1.00) = 0.8413$. The shaded area on both pictures is identical — we just relabeled the axis.

Example 4.2.1. (*One-Sided Normal Probability*) Adult human heights follow approximately $N(67, 3)$ (inches). What fraction of adults are shorter than 61 inches?

Step 1. $z = \frac{61 - 67}{3} = \frac{-6}{3} = -2.00$.

Step 2. $P(Z \leq -2.00) = 0.0228$.

About 2.3% of adults are shorter than 61 inches. The z -score of -2.00 confirms this is 2 standard deviations below the mean — rare, but not impossible.

Example 4.2.2. (*Two-Sided Normal Probability*) A machine fills juice bottles. Fill amounts follow $N(16.0, 0.3)$ ounces. Quality control rejects bottles outside the range 15.5 to 16.6 ounces. What fraction of bottles pass inspection?

Step 1. Convert both endpoints:

$$z_1 = \frac{15.5 - 16.0}{0.3} = \frac{-0.5}{0.3} \approx -1.67, \quad z_2 = \frac{16.6 - 16.0}{0.3} = \frac{0.6}{0.3} = 2.00.$$

Step 2. $P(-1.67 \leq Z \leq 2.00) = P(Z \leq 2.00) - P(Z \leq -1.67) = 0.9772 - 0.0475 = 0.9297$.

About 92.97% of bottles pass inspection.

Example 4.2.3. (*Working Backwards: Finding a Cutoff Value*) A professor grades on a curve: the top 10% of students earn an A. Exam scores follow $N(72, 8)$. What is the minimum score for an A?

We need x such that $P(X > x) = 0.10$, i.e., $P(X \leq x) = 0.90$.

Step 1. Find z^* such that $P(Z \leq z^*) = 0.90$: from the table, $z^* \approx 1.28$.

Step 2. Convert back: $x = \mu + z^* \cdot \sigma = 72 + 1.28(8) = 72 + 10.24 = 82.24$.

Students scoring above about 82 earn an A.

Common Pitfall. When working backwards (finding x from a probability), many students forget to *convert back*. Finding $z^* = 1.28$ is not the final answer if the original variable is not standard normal. Always apply $x = \mu + z^* \cdot \sigma$ at the end.

Exercises 4.2

4.2.1. IQ scores follow $N(100, 15)$. Find: (a) $P(X \leq 115)$; (b) $P(X > 85)$; (c) $P(85 \leq X \leq 115)$. Sketch a curve for each.

4.2.2. The weight of a certain brand of cereal box follows $N(20.1, 0.4)$ oz. (a) What fraction of boxes weigh less than 20 oz? (b) A box is underweight if it contains less than 19.5 oz. What fraction are underweight?

4.2.3. For $X \sim N(50, 10)$: find the value x such that $P(X \leq x) = 0.75$.

4.3 The Sampling Distribution of $\bar{x}$

So far, X has described a single randomly chosen individual. Now we shift: what happens when we take a *sample* and compute its *mean*?

4.3.1 The Idea

Imagine taking a random sample of $n = 25$ students and measuring their heights. You compute $\bar{x}$. Now take a different sample of 25 students: different $\bar{x}$. Do it 10,000 times. The 10,000 values of $\bar{x}$ form their own distribution — the **sampling distribution of $\bar{x}$** .

Definition: Sampling Distribution of $\bar{x}$. The **sampling distribution of $\bar{x}$** is the distribution of all possible sample means $\bar{x}$ that could arise from repeated sampling of size n from the same population.

Definition: Standard Error. The **standard error of the mean** is the standard deviation of the sampling distribution of $\bar{x}$:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

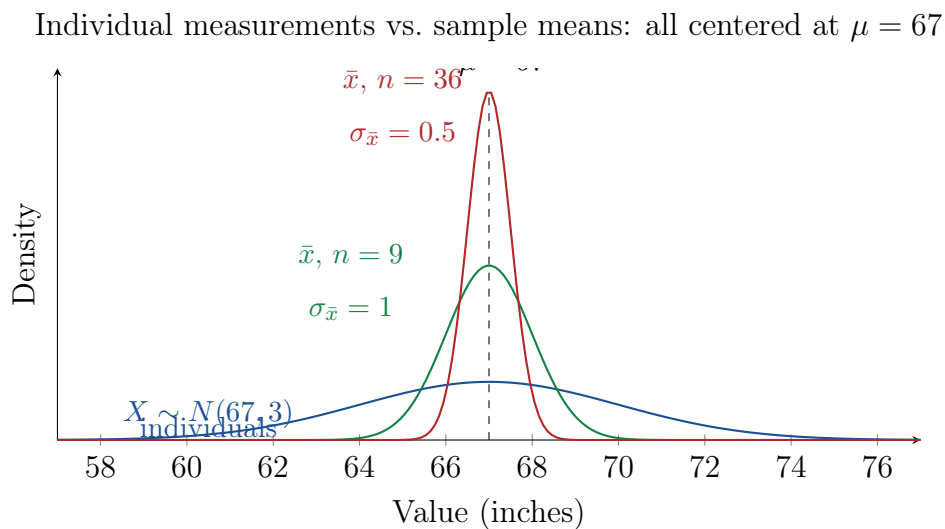
It measures how much $\bar{x}$ varies from sample to sample.

Notice the subscript: $\sigma_{\bar{x}}$ is the standard deviation of the *sample mean*, not of individual observations. They are different things.

4.3.2 Why the Standard Error Shrinks

The $\frac{1}{\sqrt{n}}$ factor is the key. As sample size n grows, $\sigma_{\bar{x}}$ shrinks. Larger samples produce more stable, less variable averages. This is the mathematical reason that bigger samples give more reliable estimates.

Visual Insight. The figure below shows the distribution of individual heights ($\sigma = 3$) alongside the sampling distribution of $\bar{x}$ for $n = 9$ and $n = 36$. All three are centered at $\mu = 67$. But the sample mean distributions are narrower by factors of $\sqrt{9} = 3$ and $\sqrt{36} = 6$.



All three curves have the same area ($= 1$). The taller and narrower the curve, the more precisely $\bar{x}$ estimates μ . Quadrupling the sample size ($9 \rightarrow 36$) only halves the standard error. You have to quadruple n to halve $\sigma_{\bar{x}}$.

Exercises 4.3

4.3.1. Heights follow $N(67, 3)$. For samples of size n , compute $\sigma_{\bar{x}}$ when $n = 4, 16, 25, 100$. What pattern do you notice?

4.3.2. A population has $\mu = 50, \sigma = 20$. How large must n be so that $\sigma_{\bar{x}} < 2$?

4.3.3. Explain in your own words: why does averaging a large sample produce a more reliable estimate of μ than averaging a small sample?

4.4 The Central Limit Theorem

We know that if the population is normal, the sample mean $\bar{x}$ is also normally distributed. The stunning fact is that this holds even when the population is not normal.

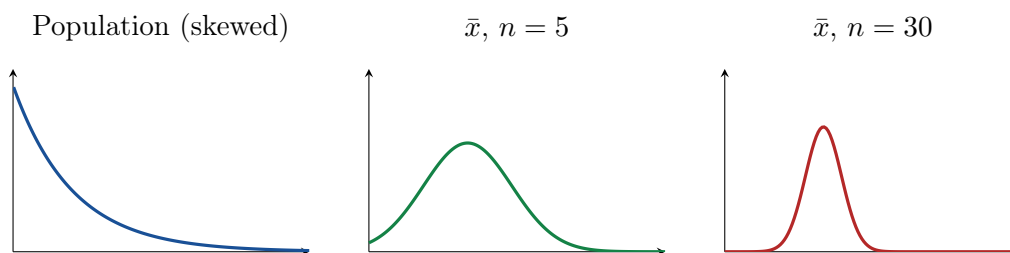
Key Formula. Central Limit Theorem (CLT). Suppose X is any random variable with mean μ and standard deviation σ . For a random sample of size n :

$$\bar{x} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

This approximation improves as n increases. A common guideline: the approximation is good for $n \geq 30$; it is excellent for $n \geq 50$.

The word “any” in the CLT deserves emphasis. The population can be skewed, bimodal, uniform — it does not matter. Average enough values together, and the distribution of the average becomes normal.

Visual Insight. The pictures below show what happens when you take samples from a strongly skewed (right-skewed exponential) population and plot the distribution of sample means.



Left: the original population is strongly skewed right. Middle: the distribution of sample means for $n = 5$ is still a bit skewed. Right: for $n = 30$, the distribution of sample means is strikingly bell-shaped, even though the original population was not. The CLT is working.

Common Pitfall. The CLT describes the distribution of *sample means* $\bar{x}$, not of individual observations x . The original data do not become normal. Individuals drawn from a skewed population remain skewed. It is their *average* that approaches

normality.

4.5 Computing Probabilities for Sample Means

Once we know $\bar{x}$ is approximately $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$, we compute probabilities exactly as in Section 4.2 — but with the standard error $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ in place of σ .

Key Formula. For a sample mean $\bar{x}$ from a population with mean μ and std. dev. σ :

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}.$$

This z -score measures how many standard errors $\bar{x}$ is from μ .

Example 4.5.1. (CLT Probability) Coffee cup fill amounts follow $N(8.1, 0.4)$ oz (or any distribution with this mean and standard deviation). A sample of $n = 16$ cups is drawn. What is the probability the sample mean fill is between 7.95 and 8.25 oz?

Step 1. Identify the sampling distribution: $\bar{x} \sim N\left(8.1, \frac{0.4}{\sqrt{16}}\right) = N(8.1, 0.10)$.

Step 2. Convert both endpoints:

$$z_1 = \frac{7.95 - 8.1}{0.10} = -1.50, \quad z_2 = \frac{8.25 - 8.1}{0.10} = 1.50.$$

Step 3. $P(-1.50 \leq Z \leq 1.50) = 0.9332 - 0.0668 = 0.8664$.

About 86.6% of samples of 16 cups will have a mean fill between 7.95 and 8.25 oz. Compare: $P(7.95 \leq X \leq 8.25)$ for a single cup is only about 57%. Averages cluster around the population mean far more tightly than individual values do.

Example 4.5.2. (Unusual Sample Mean) The math SAT follows (approximately) $N(530, 115)$. A random sample of $n = 50$ students is taken. What is the probability the sample mean exceeds 560?

Step 1. $\sigma_{\bar{x}} = 115/\sqrt{50} \approx 16.26$.

Step 2. $z = \frac{560 - 530}{16.26} \approx 1.84$.

Step 3. $P(Z > 1.84) = 1 - P(Z \leq 1.84) = 1 - 0.9671 = 0.0329$.

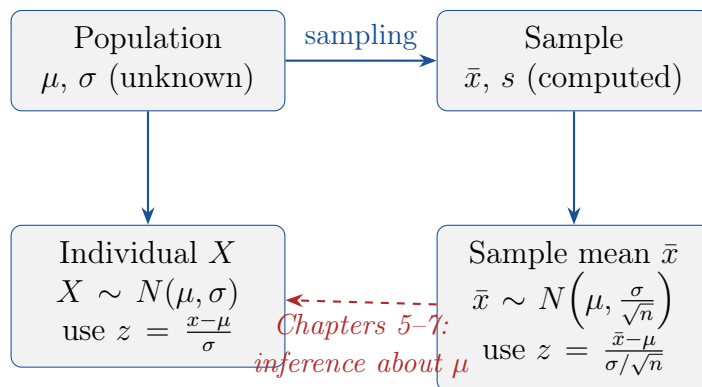
Only about 3.3% of samples of 50 students produce a mean above 560. If we actually observed $\bar{x} = 560$ from such a sample, we would have reason to question whether this sample is truly representative of the population. This reasoning — “how surprising is this sample result?” — is exactly what **hypothesis testing** formalizes in Chapter 6.

Exercises 4.5

- 4.5.1. The lifetime of a certain battery follows a distribution with $\mu = 500$ hours and $\sigma = 40$ hours. For a sample of $n = 64$ batteries: (a) What is the sampling distribution of $\bar{x}$? State mean and standard error. (b) $P(\bar{x} \leq 495)$. (c) $P(496 \leq \bar{x} \leq 504)$.
- 4.5.2. A population has $\mu = 100$, $\sigma = 25$. How would your answer to (a) change if the sample size increased from $n = 25$ to $n = 100$? Describe the effect on both $\mu_{\bar{x}}$ and $\sigma_{\bar{x}}$.
- 4.5.3. Explain why $P(\bar{x} > 535)$ is smaller than $P(X > 535)$ for the SAT population above. What does this say about using sample means to make decisions versus using individual observations?

4.6 The Big Picture: Four Weeks in One Diagram

We have now built the foundation of the entire course. Here is every idea so far in one picture:



- Chapters 1–2: We described samples $(\bar{x}, s)$ and introduced probability.
- Chapter 3: We introduced continuous probability distributions and the z -table.
- Chapter 4 (this chapter): We connected individual measurements $(N(\mu, \sigma))$ to sample means $(N(\mu, \sigma/\sqrt{n}))$ via the CLT and z -scores.
- Chapters 5–7: We will use this machinery to make formal claims about the unknown population parameters μ and p . Every test in those chapters will use a z -score of the form $\frac{(\text{estimate}) - (\text{claim})}{\text{standard error}}$.

Chapter 4 Review

1. Newborn birth weights in a certain hospital follow $N(7.5, 1.1)$ pounds. (a) What fraction of newborns weigh less than 6 pounds? (b) What fraction weigh between 7 and 9 pounds? (c) The heaviest 5% of newborns weigh more than how many pounds? (d) For samples of $n = 36$ newborns, what is the sampling distribution of $\bar{x}$? Give the mean and standard error. (e) $P(\bar{x} > 7.8)$ for the sample of 36.
2. A skewed-right distribution has $\mu = 20$, $\sigma = 8$. Samples of $n = 40$ are taken. Can we use the normal distribution to compute probabilities about $\bar{x}$? Justify your answer, then compute $P(\bar{x} \leq 19)$.
3. Two students argue: “If I take a sample of 100 people from a skewed population, the data in my sample will be approximately normally distributed.” Is this correct? Explain clearly, using the CLT.
4. *Big picture:* Write three sentences that explain why the z -score formula $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ is the bridge between descriptive statistics (Chapter 1) and inferential statistics (Chapter 5). Why is the denominator $\sigma/\sqrt{n}$ rather than just σ ?